

Ganesh Gupta

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EXPERIENCE

Oracle | *Software Engineer, AI Platform*

Jun 2025 – Present

- Engineered a cloud-native **distributed executor** routing pipeline nodes to **Docker** containers and serverless functions via typed **REST APIs**, implementing **OAuth-based auth** and **RBAC** for secure service access.
- Built a deployment-gated **evaluation framework** with **guardrails** for safe handling of model outputs, owning **design reviews** and **production debugging** and participating in **code reviews**.
- Built fault-tolerant **GPU/HPC scheduling** across **AWS and GCP**, maintaining fleet utilization through **node loss and provider hiccups** with **Go and Python** placement and resource-allocation logic.
- Backed the distributed executor with an **S3** artifact store for content-addressed memoization, sustaining **sub-200ms p99 latency** under production load.

QureAI | *ML Infrastructure Engineer*

Mar 2026 – Present

- Cut **p99 latency from 420ms to 140ms** across **50K+ daily requests** by rebuilding synchronous LLM serving into an **async, autoscaled** architecture on **GCP Cloud Run**, with **automated CI/CD** integration.
- Triaged and resolved **distributed system incidents** end-to-end on an on-call rotation for **Live Site** reliability, with automated **CI/CD rollback** and **failure alerting**.
- Built a production **observability platform** tracking **latency**, drift, and regressions across deployments, using **telemetry** to drive automated rollback decisions.

Nomura Research Institute Financial Technologies | *Software Engineer*

Sep 2019 – Aug 2023

- Operated **25+ Java and Python microservices** serving **1M+ users** at **99.9% uptime** over 4 years on **AWS (EC2, RDS, SQS)**, owning **system design and operations**.
- Cut **production escalations 60%** in one quarter by standardizing **retry, timeout, and error-propagation** contracts across **distributed services**, and drove **root cause analysis** and **post-incident reviews**.
- Cut a nightly SQL aggregation pipeline from **8 minutes to 90 seconds** by moving to **indexed views** and **batch partitioning**, eliminating row-level lock contention.
- Led **architecture and code reviews** across production services, evaluating scalability, reliability, testability, and operational tradeoffs while mentoring engineers on **system design** and **reliability**.

PROJECTS

VPC Control Plane Simulator

github.com/ganeshhgupta/vpc-control-plane-sim

- Built an AWS-style **VPC networking** simulator with **CIDR validation** and overlap detection, **longest-prefix-match routing**, stateful **security group** and stateless **NACL** (access control) evaluation, and **drift detection**, on **Next.js** with **PostgreSQL**.

Watch-your-LLM

github.com/ganeshhgupta/watch-your-llm

- Built a self-hosted **observability** platform that traces requests, **latency**, and cost through a Python SDK decorator, sending traces asynchronously to a **FastAPI** collector on **PostgreSQL**.

Distributed KV Store

github.com/ganeshhgupta/Distributed-KV-Store

- Built a distributed KV store simulator with **quorum-based replication**, **write-ahead logging**, and key **partitioning** across a 3-node **leader/follower** cluster, with request traces for inspecting each operation's path.

EDUCATION

University of Texas at Arlington | *M.S. Computer Science, GPA 4.0/4.0*

Aug 2023 – May 2025

MCKV Institute of Engineering | *B.Tech Computer Science*

Aug 2016 – Jul 2020

SKILLS

Languages: Go, Python, C++, Java, SQL

Secure Systems: OAuth, RBAC, Access Control, Guardrails, Security Groups, NACLs

Distributed Systems: Distributed Scheduling, Fault-Tolerant Placement, Consensus, Replication

Cloud & DevOps: AWS, GCP, Docker, CI/CD, Observability, Telemetry, Automated Testing

Agentic AI: OpenAI Agents SDK, CrewAI, AutoGen, MCP

Certifications: AWS Certified Developer Associate